15. How many zeros are there in 100^{10} ?

(i) 2 (ii) 3 (iii) 10 (iv) 20

16. $2^{40}+2^{40}$ is equal to

(i) 4^{40} (ii) 2^{80} (iii) 2^{41} (iv) 4^{80}

3.4 Unit Digit of Numbers in Exponential Form

Manipulating with exponent is very interesting and funny too.

We know that $9^3=9\times9\times9=729$, thus the unit digit (the last number of expanded form) of 9^3 is 9. Similarly, 4^4 is $4\times4\times4\times4=256$. Thus the unit digit of 4^4 is 6.

Can you guess the unit digit of 230^{116}, 181^{47}, 55^4, 56^{20} and 9^{29} ?

It is very difficult to find by expanding the exponential form. But, we can try to tell the unit digit by observing some patterns.

Look at the following number pattern.

$10^3=10\times10\times10=1000$

$10^4=10\times10\times10\times10=10000$

$10^5=10\times10\times10\times10\times10=100000$

Thus, multiplying 10 by itself several times, we always get the unit digit as 0. In other words, 10 raised to the power of any number has the unit digit 0. That is, the unit digit of 10^x is always 0, for any positive integer x.

This is also true when the base is multiples of 10. Consider,

$$40^2=(4\times10)^2=4^2\times10^2$$
$$=16\times100=1600$$

Similarly, $230^{116}=(23\times10)^{116}=23^{116}\times10^{116}$

Thus, the unit digit of 230^{116} is 0.

Now, observe that,

$$1^5=1\times1\times1\times1\times1=1$$
$$1^6=1\times1\times1\times1\times1\times1=1$$

We learnt to expand 11 as 10+1.

So, $(10+1)^2=11^2=11\times11=121$

Similarly, $131=130+1=(13\times10)+1$

$[(13\times10)+1]^2=131^2=131\times131=17161$

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